

The Effect of Covariate Incompleteness on CATE Estimation with CRM data

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Submitted on 15 June 2026

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Abstract

Background. Organisations increasingly apply causal machine learning to estimate Conditional Average Treatment Effects (CATE) from CRM data to support individualised targeting decisions. However, CRM data frequently suffers from incomplete covariate information arising from inconsistent data collection practices — a data quality problem that has received limited attention in the causal machine learning literature. This thesis investigates how covariate incompleteness affects the accuracy, stability, and ranking quality of CATE estimation, and whether the choice of missing-data handling strategy moderates these effects.

Methods. A simulation study is conducted using a synthetic CRM population calibrated to a real B2B quote-to-order dataset. Following the MIDA framework, a factorial design crosses four missingness levels (0%, 10%, 50%, 70% MAR), three handling strategies (complete-case analysis, mean/mode imputation, and MICE), and four estimators (S-learner, T-learner, X-learner, Causal Forest), producing 40 estimation pipelines evaluated across $R = 1000$ replications. Performance is assessed using RMSE, replication-level standard deviation of RMSE, and top-20% targeting overlap against known true individual treatment effects.

Results. Estimator choice dominates, accounting for the overwhelming majority of variance in RMSE ($SS = 47.94$) compared to missingness level ($SS = 1.66$) and imputation method ($SS = 0.89$). The S-learner and Causal Forest consistently outperform the T-learner and X-learner across all conditions. Missing-data handling matters primarily at high missingness (50%-70%), where complete-case analysis dramatically degrades accuracy and stability, while MICE and mean/mode imputation preserve near-baseline performance by retaining the full sample. In term of ranking quality the the S-learner and Causal Forest achieve top-20% targeting overlap of approximately 0.35-0.40 under complete data.

Conclusion. Practitioners should prioritise estimator selection over imputation sophistication, and any imputation strategy preserving sample size is preferable to listwise deletion at high missingness. The results further highlight that the practical value of CATE estimation in CRM environments is fundamentally bounded by the signal-to-noise ratio of individual-level treatment effect heterogeneity, independently of the analytical pipeline employed.

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1 Introduction

In a business analytics workshop class, one student team enthusiastically presented a recommendation to a company: "Additional employees should be scheduled to be hired in August, because the data suggests that quote requests in August skyrockets". The recommendation appeared actionable, data-driven, and operationally relevant. Yet the other team reached a different conclusion. Looking more closely at the same data, they observed that there were particular 400 quote requests in August that had occurred on the same day and more peculiarly generated at the exact same time. Their message to the company was therefore not to hire more employees, but to audit their data collection process.

This instance illustrates a problem in analytics-driven decision-making. Recommendations can be produced even when the underlying data consists of recording errors or missing information and if not critically evaluated can be completely unreasonable. Recent industry evidence highlights the substantial economic consequences of poor data quality. According to the IBM Institute for Business Value, more than one quarter of organizations estimate annual losses exceeding USD 5 million due to poor data quality, while 7% report losses above USD 25 million annually (Krantz and Jonker, 2026). Concerns regarding poor data quality are not new. Prior to the widespread adoption of modern data analytics and explosion of Big Data, Redman (1998) already emphasized that low-quality data substantially increases operational costs and estimated that related inefficiencies may account for 40-60% of organizational expenses.

As organizations increasingly seek to extract value from growing volumes of customer and operational data, analytical methods have evolved to support more sophisticated forms of decision-making (Sharafuddin and Belik, 2022). Firms like Booking.com and Vinted are leaders in developing culture of experimentation in the organization and reap the benefits of it (Thomke, 2024). In particular, machine learning and causal inference methods are increasingly applied in CRM context to evaluate customer communication strategies, optimize targeting decisions, and personalize interventions (Lemmens et al., 2025).

At the same time, CRM-intensive environments frequently operate under substantial data quality limitations. As a result, organizations may accumulate incomplete or unreliable customer information but still be driven by benefits they expect from implementation of analytical methods. Although data quality challenges were documented in early 2000s Andrea and Miriam (2005), they remain prevalent today despite substantial advances in data infrastructure, analytics, and big data technologies.

These challenges are particularly important within causal machine learning settings because Conditional Average Treatment Effect (CATE) estimation depends on observed covariate information (Baiardi and Naghi, 2024). Consequently, incomplete covariate in-

formation may substantially distort individualized treatment effect estimates and reduce the reliability of customer-level targeting decisions (Baiardi and Naghi, 2024).

As a result, the idea for this thesis emerged from discussion and intersection of these topics that combine fields of data quality management, causal inference, machine learning and marketing.

1.1 Problem Statement

Operational business decisions are increasingly based on analytical estimates derived from CRM data, but when customer information is incomplete, it remains unclear how missing-data handling strategies affect the reliability of CATE estimates and the managerial decisions based on them (ranking).

Operational business decisions are increasingly informed by analytical models whose reliability depends on the quality of the underlying data. When data quality is compromised, the validity of the resulting estimates and the decisions based on them becomes uncertain. This creates a managerial dilemma of how much and should organizations invest in improving data quality to support reliable and analytically sound decision-making.

Organizations increasingly use analytical approaches to support customer-level decision-making. In CRM-intensive sales environments, such as construction materials, window frame manufacturing, and consultative service provision, retail, firms may use conversion prediction, churn modelling, Conditional Average Treatment Effect (CATE) estimation to decide which customers should receive follow-up calls, reminders, discounts, or other commercial interventions. These methods are attractive because they promise more targeted and data-driven allocation of sales and marketing resources.

However, analytical models can only be as valuable as the data on which they depend. Value of data depends on the quality of the processes, systems, and procedures through which they are generated, stored, and maintained (Haug et al., 2011, p. 169). In the context of CRM analytics, this means that sophisticated models may still produce unreliable outputs if the customer information used as input is incomplete, inconsistent, or inaccurately recorded. In CRM-intensive sales processes, where customer information is often created through a combination of manual data entry, sales interactions, and digital system integration. Sales associates may record customer characteristics, communication history, quote information, and transaction details across multiple follow-up stages. In practice, these data collection procedures are not always standardized, consistently followed, or systematically audited. Moreover, when customer information is integrated across multiple platforms, differences in system architecture, recording practices, and validation procedures may introduce additional inconsistencies. As a result, organizations may accumulate CRM datasets without full visibility into how the data were originally generated or why some information is missing.

One important dimension of this broader data quality problem is completeness. Missing covariate information may arise from the data-generating process itself: sales employees may skip fields, certain customer attributes may only be recorded for new customers, or specific information may be unavailable at the time of quote creation. Consequently, missingness is not only a technical inconvenience but also a reflection of organizational data collection practices. Depending on the underlying missingness mechanism, the absence of information and consequently way we decide to deal with it effects model training, treatment effect estimation, and the resulting ranking of customers for targeted interventions (Berrevoets et al., 2023; Chen and Cummings, 2025).

This creates both a managerial and methodological dilemma. Organizations can respond to incomplete CRM data by investing in better data collection, process standardization, auditing, and system integration. However, such improvements require organizational resources. Alternatively, analysts attempt to recover information statistically through missing-data handling strategies. The selected missing-data handling method may directly influence CATE estimates and, consequently, the business decisions derived from them (Berrevoets et al., 2023).

1.2 Research Objective

This aim of this thesis is to investigate how Conditional Average Treatment Effect (CATE) estimation changes under varying levels of missing covariate information, different missing-data handling techniques, and different estimator choices. Specifically, it examines how covariate incompleteness in CRM data affects the accuracy and stability of CATE estimates given a certain scenario.

1.3 Research Question

“How does covariate incompleteness in CRM data, together with the choice of missing-data handling strategy and estimator, affect (1) the accuracy, (2) stability of Conditional Average Treatment Effect (CATE) estimation and (3) the quality of resulting ranking-based targeting decisions?”

1.4 Academic Relevance

This research contributes to several interconnected literature streams, including causal machine learning, uplift modeling in marketing analytics, missing data theory, and data quality management. While prior research has extensively examined missing data problems and causal inference methods separately, limited attention has been given to how incomplete covariate information affects the robustness of Conditional Average Treatment Effect (CATE) estimation in CRM context.

A particular research gap exists in connecting methodological developments originating from statistics and causal inference literature with operational CRM data collection problems commonly observed in marketing and sales environments. This thesis therefore contributes by extending methodological insights from causal inference and missing data literature into applied context.

The study further contributes through the application and comparison of multiple causal meta-learning approaches for heterogeneous treatment effect estimation, including S-learners, T-learners, X-learners, and causal forests (Künzel et al., 2019) (Wager and Athey, 2018). By evaluating model performance under varying handling techniques and levels of covariate incompleteness, the research contributes to broader discussions regarding the consistency of causal machine learning methods under imperfect data conditions.

1.5 Managerial Relevance

The key managerial challenge is not merely predictive performance, but the reliability of decision-making under incomplete data conditions. Organizations may implement sophisticated analytical models while underestimating the extent to which incomplete customer information can distort treatment effect estimates. The problem directly affects several stakeholder groups. Sales managers and sales associates are responsible for recording customer interactions and maintaining CRM systems, while marketing managers evaluate the effectiveness of communication and intervention strategies. In addition, senior management is concerned with return on investment, allocation of sales resources, and the financial consequences of unreliable analytical outputs.

This study seeks to help firms evaluate misallocation of sales effort, continuation or termination of communication strategies based. In addition, the demonstration seeks to turn attention to the trade-off between investing in improved data quality practices and accepting the risks associated with unreliable targeting and decision-making systems.

In addition, this study aims to draw attention to the role of data quality education within business analytics and data science programmes as key settings in which future practitioners acquire the foundational knowledge for their professional roles. While analytical modelling techniques are often taught in considerable depth, the processes through which data are generated, maintained, and compromised may receive less systematic attention. This concern has also been raised by David J. Hand, Emeritus Professor of Mathematics and Senior Research Investigator at Imperial College London, who has argued that data quality and “dark data” deserve greater attention in analytics education and practice (unSILOed Podcast with Greg LaBlanc, 2022).

1.6 Research Design Overview

This study follows a simulation-based research design using synthetic CRM data. The synthetic population is generated to reflect characteristics observed in real B2B CRM quote-to-order data. The design is structured according to the MIDA framework (Blair et al., 2023) and results are derived from 1000 simulation replications.

1.7 Thesis Structure

Thesis is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the methodology, structured according to the MIDA framework: the synthetic population level CRM data-generation, the inquiry, the data strategy including the missingness, and the answer strategy comprising handling methods, the four CATE estimators and performance measures. Section 4 presents the simulation results across the three research questions: CATE accuracy, stability, and ranking quality for targeting decisions. Section 5 discusses the findings in relation to the existing literature and derives theoretical and managerial implications. Section 6 concludes with a summary of main findings, limitations, and directions for future research.

2 Literature Review

The aim is to introduce the key concepts and overview of studies exploring related research question. First, interconnected areas are covered, namely, data quality management with a focus on completeness, missing data mechanisms, causal inference and CATE estimation. The intersection of causal machine learning with incomplete data subsection aims to provide an overview of what similar simulation studies have found.

2.1 Data Quality Management

In modern business environments, data has increasingly become recognized as a strategic organizational resource. In 2006, David Humby famously stated that “data is the new oil” (O’Neil et al., 2023, p. 1), emphasizing the growing economic value of data-driven decision making.

As with any strategic resource, data requires management and quality control in order to maximize its value. Within organizations, the primary objective of data quality management is to ensure that data used for operational and strategic decision-making is accurate, reliable, and usable (Keller et al., 2017, p. 89). Poor quality data may distort business intelligence, weaken predictive models, and ultimately lead to inefficient allocation of organizational resources. However, Thomas Redman (Redman, 1998) extended discussion from purely monetary cost to stating that consequences carry negative effects even on organizational morale and environment (Redman, 1998, p.80).

2.1.1 Data Quality Dimensions: Completeness

Data quality is commonly understood as a multidimensional concept. The DAMA-DMBOK defines data quality as the “degree to which dimensions of data quality meet requirements” (DAMA, 2014, p. 422). From a practical data management perspective, the DAMA-DMBOK framework identifies several key dimensions of data quality, including validity, completeness, consistency, integrity, timeliness, currency, reasonableness, uniqueness, and accuracy.

Although the DAMA-DMBOK is considered one of the leading practitioner-oriented frameworks in data management, its foundations are built upon earlier contributions by pioneers in the field such as Redman (1996), Wang and Strong (1996), and English (1999). Following this tradition, the present thesis adopts the DAMA-DMBOK framework as its primary conceptualization of data quality.

Among the identified dimensions, this study focuses specifically on completeness. Completeness is defined as the extent to which required data is present within the dataset (DAMA, 2014, p. 428). In CRM, completeness problems frequently emerge due to operational realities such as manual data entry, inconsistent follow-up documentation, selective recording of customer interactions, and fragmented information systems.

The completeness dimension becomes particularly important in causal estimation settings because causal identification relies heavily on the availability of observed covariate information. This is explored in depth in Section 2.3 .

2.2 Missing Data Theory

Missing Data Mechanisms Data may be absent for various operational, behavioral, or structural reasons. Historically, missing data was often treated as a secondary technical nuisance rather than a central methodological problem. However, Donald B Rubin (1976) formalized the statistical treatment of missing data mechanisms and demonstrated that assumptions regarding missingness fundamentally affect statistical inference. In his work, Rubin discusses situations in which data is missing at random (MAR) and argues that likelihood-based and Bayesian inference procedures are less sensitive under MAR assumptions.

Later literature formalized the commonly used taxonomy of missing data mechanisms into Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) (Little and Donald B.. Rubin, 2020, pp. 13–23).

Under MCAR, missingness occurs independently of both observed and unobserved variables. In CRM settings, this could occur due to random technical failures or accidental loss of records. Under MAR, missingness depends on observed information. For example, a particular salesman may have weakest discipline in data recording as a result, accounts managed by that salesperson will tend to have more incomplete data in certain fields.

Finally, under MNAR, missingness depends on the unobserved value itself. In CRM environments, customers with lower purchase intent or negative interactions may be less likely to provide complete information, making missingness structurally related to the outcome of interest.

While MCAR assumptions simplify statistical treatment of missingness, they are rarely realistic in CRM environments where data generation is often influenced by customer profile or operational activities. As a result, MAR and MNAR mechanisms are more realistic in CRM data generation.

2.3 Causal Inference

Determining causality has become a focal task in marketing analytics because it provides a direct answer to questions such as: what actions lead customers to purchase, retain, or respond to marketing efforts? Controlled experiments, commonly referred to as A/B tests, are among the most effective tools for answering such questions (Ron et al., 2007). The metric that shows effect of chosen effort or strategy on an individual level is The Conditional Average Treatment Effect.

CATE and Incomplete Covariates The Conditional Average Treatment Effect (CATE) focuses on estimating treatment effects conditional on observed individual characteristics. CATE builds on the potential outcomes framework and is formalized within Rubin’s causal model (Imbens and Donald B.. Rubin, 2015).

CATE (individual level) in the context of an A/B test can be expressed as (Morgan and Winship, 2015, p.47):

$$\tau(x_i) = \mathbb{P}(Y_i = 1 \mid W_i = 1, X = x_i) - \mathbb{P}(Y_i = 1 \mid W_i = 0, X = x_i)$$

CATE estimation is inherently sensitive to missing covariate information because it relies on conditioning on observed characteristics ($X=x$) to approximate counterfactual outcomes. This becomes problematic when relevant covariates are unavailable, as it weakens the ability to meaningfully compare treated and control units with similar characteristics (Morgan and Winship, 2015, p.46).

2.4 Causal Machine Learning and Missing Data Handling

Traditional causal inference methods often struggle in high-dimensional environments characterized by complex interactions, nonlinear relationships, and heterogeneous treatment effects. CRM environments frequently exhibit these characteristics due to the large number of customer attributes, behavioral indicators, and operational variables involved in customer interactions.

Recent developments in causal machine learning attempt to address these limitations by combining predictive machine learning methods with causal inference frameworks.

As discussed by Curth et al. (2024), machine learning methods provide opportunities to estimate individualized treatment effects while flexibly modeling nonlinearities and heterogeneity. However, in high-dimensional settings, flexible learners may amplify noise, overfit sparse patterns, or become highly sensitive to data quality limitations (Curth et al., 2024). To extend on Curth et al. (2024) discussion, a simulation study done by Jacob (2021) compares meta-learners that were designed estimating CATE directly (Künzel et al., 2019) and machine learning methods that were adjusted to CATE estimation context (Wager and Athey, 2018). The following Table 1 summarizes characteristics of methods of interest in this study.

Table 1: Comparison of CATE Estimators: S-learner, T-learner, X-learner, and Causal Forest

Criterion	S-learner	T-learner	X-learner	Causal Forest
Category	Meta-learner	Meta-learner	Meta-learner	Modified ML method
Core idea	Single model with treatment included as a covariate	Two separate outcome models, one per treatment arm	Imputes individual treatment effects using the opposite group's fitted model	Honest tree ensemble whose splitting criterion directly targets treatment effect heterogeneity
Number of models	1 outcome model (2 predictions)	2 outcome models	3 outcome models + 2 CATE models (5 total)	1 causal forest + 2 nuisance functions
CATE estimator	$\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$	$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$	$\hat{\tau}(x) = g(x)\hat{\tau}_0(x) + \{1-g(x)\}\hat{\tau}_1(x)$	Weighted local moment estimator via forest weights $\alpha_i(x)$
Optimisation target	Predictive accuracy of Y	Predictive accuracy of Y per arm	Predictive accuracy of imputed treatment effects	Directly targets heterogeneity in $\tau(x)$

Continued on next page

Table 1 – *Continued*

Criterion	S-learner	T-learner	X-learner	Causal Forest
Key strength	Simple; leverages all data in one model	Intuitive; flexible choice of base learner	Efficient under imbalanced treatment arms; reduces bias relative to T-learner	Asymptotically valid confidence intervals; honest estimation
Key weakness	Regularisation may shrink the treatment effect toward zero; treatment indicator may be ignored	Optimises prediction not treatment effect; can diverge when arms shrink different covariates	More complex pipeline	Computationally intensive

Jacob (2021) finds that in simulation studies the Causal Forest performs competitively, particularly under non-linear treatment effect, while meta-learners can be more accurate when the underlying functions are relatively simple. However, an important consideration is that no single method dominates in all settings, and that comparing estimates across multiple estimators provide a robustness check (Künzel et al., 2019) (Jacob, 2021).

In terms of missing data handling methods, the amount of available techniques in the literature is extensive (Zhou et al., 2024). The present study therefore evaluates three handling strategies chosen to reflect a spectrum of analytical practice, ordered by increasing awareness of the missingness mechanism and its implications. A practitioner with no explicit consideration of missingness would default to Complete Case Analysis (CCA), discarding all incomplete records. A practitioner aware that imputation is preferable but without consideration of the underlying mechanism might resort to mean and mode imputation which is a simple and widely available approach that is, however, only valid under MCAR (Zhou et al., 2024). Finally, a practitioner who explicitly recognises the MAR mechanism and its implications would turn to Multiple Imputation by Chained Equations (MICE), developed specifically to handle MAR data by preserving the uncertainty and dependence structure among covariates (Buuren, 2018).

The interaction between missing data handling and machine learning model performance has begun to attract research attention, though primarily in predictive rather than causal settings. **baysalerez2024** demonstrate across multiple missingness levels (10%, 50%, and 70%) that the choice of imputation method meaningfully affects both predictive performance and model explainability, concluding that imputer and classifier selection should be treated as a joint rather than sequential decision. Crucially, their results show that mean imputation consistently produced the largest distortions in model explanations, reinforcing the limitation that simple single-value imputation is inadequate when missingness departs from MCAR (Zhou et al., 2024).

3 Methodology

This thesis is a quantitative simulation study on synthetic data generated based on CRM environment data in quote-to-order context of a Dutch company studied in the past. The objective is not to reproduce one empirical CRM dataset exactly, but to create a controlled synthetic environment in which the true CATE is known. This allows the performance of different analytical pipelines to be evaluated under varied missingness conditions.

The simulation is defined and implemented using the MIDA framework (Blair et al., 2023), which provides a formal architecture for defining and executing simulation studies. The simulation follows a factorial design in which multiple design factors are systematically combined: missing-data worlds, missing-data handling strategies, and CATE

estimation methods (Parke, 2010). Simulation methods are particularly suitable for this research because they allow to define and clearly see the data generating mechanisms, control levels of covariate incompleteness while preserving knowledge of the true underlying treatment effect (τ_i). Unlike observational CRM data, where true individual-level treatment effects are unobserved, simulated environments make it possible to evaluate estimation accuracy and variance under known data-generating conditions (Morris et al., 2019).

3.1 Research Design

The MIDA framework is used to describe all elements of the study of model, inquiry, data strategy, and answer strategy (Blair et al., 2023). In this study, in the *model* is where the outcome variables and probabilities, true treatment effect, covariates are defined. The *inquiry* concerns what is estimated, in this case it is CATE. The *data strategy* consists of explaining missigness mechanism. The *answer strategy* are the estimators and imputation methods used.

Setting of the quote-to-order context Potential customers request quotations for doors manufacturing services by submitting project information, such as dimensions, material preferences, delivery requirements, and other order specifications. Sales associates review these requests, prepare quotations, and may communicate further before the final quote value is determined. Customer then decide whether to reject the offer, or confirm the order by paying an initial deposit. The causal mechanism of interest is whether follow-up communication after the final quote determination increases the probability of conversion, and whether this effect differs across customer and quoted value.

3.1.1 MIDA: Model

A model element is defined as a set of *reference models*, where each model represents how the world works and what events take place (Blair et al., 2023).

Each unit i represents one quote request. For each quote request, the data-generating process produces a treatment indicator (W_i), two potential outcomes ($Y_i(0)$ and $Y_i(1)$), their underlying conversion probabilities (p_{0i} and p_{1i}), and the true individual-level treatment effect (τ_i). A synthetic population of $N = 50,000$ units is generated once; samples of $n = 2000$ are drawn from this population in each replicate.

Covariates. The covariate set is informed by prior experience with a B2B quote-to-order CRM dataset and reflects the key drivers of conversion in that context. Ten variables are generated: three categorical variables (DEAL_OWNER, CHANNEL_TYPE, RFM_SEGMENT), two count variables (NUM_PREVIOUS_ORDERS, NUM_PRODUCT), and five continuous variables (CUSTOMER_LIFETIME_VALUE_LOG, QUOTE_VALUE,

DAYS_TO_SEND_QUOTE, QUOTE_REQUEST_MONTH, DAYS_BETWEEN_FIRST_AND_LAST_QUOTE_LOG).

Baseline conversion probability: p_{0i} . The untreated potential outcome probability is defined as

$$p_{0i} = \Pr(Y_i(0) = 1 \mid \mathbf{X}_i).$$

To obtain baseline conversion mechanism, p_{0i} is parameterised using a logistic surrogate model (Molnar, 2020) of a previously trained LightGBM classifier. The surrogate is a quasi-binomial GLM fitted to the LightGBM predicted probabilities, yielding an interpretable approximation of the learned conversion surface (correlation with LightGBM predictions: $r = 0.79$; RMSE = 0.12). The functional form is

$$p_{0i} = \text{logit}^{-1} \left(\beta_0 + \sum_{k=1}^K \beta_k X_{ik} \right),$$

where β_0 is calibrated so that the population-average conversion rate equals the observed rate of 45%. The surrogate coefficients are reported in Appendix ??.

Potential outcome under no treatment: $Y_i(0)$.

$$Y_i(0) \sim \text{Bernoulli}(p_{0i}).$$

Since each quote either converts or it does not, the binary outcome under no treatment is drawn from a Bernoulli distribution (Monroe, 2017). Given a customer’s conversion probability, the customer converts with probability p_{0i} or does not with probability $1 - p_{0i}$. It is important to note that two customers with identical covariates and therefore identical p_{0i} will not always make the same decision, hence the modelling through a random Bernoulli variable.

CATE (or individual treatment effect) $\tau(x)_i$. Treatment responsiveness is assumed to depend on quote value, RFM segment, and deal owner, reflecting the overall importance of the deal to both the customer and the company, customer relationship strength, and follow-up execution quality.

$$\tau(x)_i^{\text{logit}} = \text{clip}_{[0, 0.40]} \left(0.20 + 0.10 \tilde{z}_i + \gamma_i^{\text{RFM}} + \gamma_i^{\text{DEAL-OWNER}} \right),$$

The individual treatment effect is modelled as a constant increase in the log-odds of conversion, so that the treated conversion probability can given by $p_{1i} = \text{logistic}(\text{logit}(p_{0i}) + \tau_i^{\text{logit}})$, ensuring that $p_{1i} \in [0, 1]$ by construction.

The term $0.10 \tilde{z}_i$ introduces heterogeneity by quote value, where $\tilde{z}_i$ is the standardised log quote value. Customers with higher-value quotes are assumed to be more responsive to follow-up, as larger orders signal greater deliberation on the customer’s part and stronger incentive for the sales representative to execute the follow-up effectively.

γ_i^{RFM} captures heterogeneity across RFM segments. Lower-engagement customers are assigned larger positive weights, reflecting the assumption that highly loyal customers already have a high baseline conversion probability. $\gamma_i^{\text{DEAL-OWNER}}$ captures heterogeneity across deal owners, reflecting realistic variation in follow-up execution quality across sales representatives. The full weight tables are reported in Appendix B.

Finally, the clip to $[0, 0.40]$ serves two purposes. The lower bound of zero enforces the assumption that follow-up is never harmful. The upper bound limits the maximum individual treatment effect to approximately 10 percentage points on the probability scale for a customer at the 45% baseline, a range that deliberately exceeds the average treatment effects of 0.01 to 4.54 percentage points documented across real-world marketing datasets by Rößler and Schoder (2022), in order to introduce sufficient heterogeneity in generated treatment effects.

Treated conversion probability and potential outcome: $p_{1i}, Y_i(1)$. The treated conversion probability is obtained by applying the individual treatment effect on the log-odds of the baseline conversion probability,

$$p_{1i} = \text{logistic}(\text{logit}(p_{0i}) + \tau_i^{\text{logit}}),$$

which is equivalent to

$$p_{1i} = \text{logistic}\left(\beta_0 + \sum_{k=1}^K \beta_k X_{ik} + \tau_i^{\text{logit}}\right).$$

The treated potential outcome is then drawn as

$$Y_i(1) \sim \text{Bernoulli}(p_{1i}),$$

for the same reason as $Y_i(0)$, the quote either converts or it does not, and p_{1i} governs that probability under treatment condition.

The true individual treatment effect on the probability scale, which serves as the ground truth in all performance evaluations, is then defined as

$$\tau(x)_i = p_{1i} - p_{0i}.$$

Treatment assignment. Treatment is assigned mechanism is a Bernoulli Trial , in other words each customer is independently and randomly assigned to the treatment group with probability 0.5 (Imbens and Donald B.. Rubin, 2015, pp.48):

$$T_i \sim \text{Bernoulli}(0.5).$$

3.1.2 MIDA: Inquiry

Inquiry is defined as the research question presented in the bounds of the model (M) (Blair et al., 2023). In case of the presented Model definition, the research question is: "What the Conditional average treatment effect for each individual ($\tau(x)_i$) ?".

3.1.3 MIDA: Data Generation

Data strategy is described as a way we gather the data from the world (Blair et al., 2023).

Missing-data mechanism. After the population CRM dataset is generated, missingness is imposed on two covariates: the quantitative variable QUOTE_VALUE and the categorical variable RFM_SEGMENT.

The missing-data mechanism is MAR by construction because the probability of missingness depends on observed CRM variables (Little and Donald B.. Rubin, 2020).

For QUOTE_VALUE, missingness is generated from the observed combination of CHANNEL_TYPE and DEAL_OWNER. Specifically, quotes coming from Channel D (phone) and handled by a DEAL OWNER A are assumed to be more likely to have incomplete quote-value information. This reflects a plausible CRM workflow in which phone requests require more manual note-taking and recording completeness differs across sales associates.

For RFM_SEGMENT, missingness depends on observed customer-history information. Customers with no prior purchase history(NUM_PREVIOUS_ORDERS = 0), do not have enough behavioural information for the CRM system to assign a stable RFM segment.

The missingness worlds differ only in the severity of missingness. The complete-data benchmark m_0 contains no missing values. Worlds m_1 , m_2 , and m_3 impose increasing levels of MAR missingness. In the implementation, the missingness level refers to the target rate of missingness among records affected by missingness mechanism.

Table 3.1.3 summarizes the realities of the reference models.

Table 2: Missing-data worlds and MAR mechanisms in the CRM simulation design

World	Missingness level	Variables affected	CRM MAR mechanism
m_0	0%	None	Complete-data benchmark. No covariate values are made missing. This world represents the ideal CRM data environment.
m_1	20% MAR	Quantitative:	Missingness is generated as a function of observed CRM process variables. For QUOTE VALUE, missingness depends on the observed combination of CHANNEL TYPE and DEAL OWNER: where requests coming from CHANNEL_TYPE_D handled by a DEAL_OWNER_A are assumed to be more likely to have incomplete quote information. For RFM SEGMENT, missingness depends on customer history, captured by low number NUM_PREVIOUS_ORDERS. Customers with no prior purchase history do not have enough behavioral information for the CRM system.
m_2	50% MAR	QUOTE_VALUE.	
m_3	70% MAR	Categorical: RFM_SEGMENT.	

3.1.4 MIDA: Answer strategy

The answer strategy defines the methods used to analyse and interpret the gathered data. It encompasses data preprocessing, estimation, and the production of performance metrics and visualisations (Blair et al., 2023).

The answer strategy in this study is organised into 2 components: the missing-data handling methods, the causal machine learning estimators used to estimate heterogeneous treatment effects.

Handling Missing Data Three missing-data handling strategies are compared, ranging from the simplest possible approach to multiple imputation. All strategies are implemented with train-test separation: imputation statistics are estimated exclusively from training observations and subsequently applied to both training and test observations. This prevents information leakage from the test set into the imputation model, preserving the integrity of out-of-sample evaluation (James et al., 2017).

Complete-Case Analysis (CCA) Complete-case analysis retains only observations with no missing values on QUOTE_VALUE or RFM_SEGMENT, discarding all incomplete rows. CCA is the simplest approach in practice. However, at high missingness levels the resulting reduction in effective sample size may substantially inflate estimation variance and introduce bias if the missingness pattern is related to the outcome (Little and Donald B.. Rubin, 2020).

Mean and Mode Imputation Missing values in the continuous variable QUOTE_VALUE are replaced with the training-set mean, and missing values in the categorical variable RFM_SEGMENT are replaced with the training-set mode.

This choice behind including this imputation method is based on the fact that it is routinely applied as defaults across academic and industry data pipelines, often without further scrutiny, and are the approach most students (future practitioners) are trained to reach for first. However, the statistical literature cautions against their uncritical use (Buuren, 2018; Zhou et al., 2024). It artificially reduce the variance of imputed variables and distorts correlations between covariates, since all imputed units receive an identical value irrespective of their individual covariate profile (Little and Donald B.. Rubin, 2020). These risks are particularly pronounced when the missingness mechanism is not completely at random.

Multiple Imputation by Chained Equations (MICE). MICE is included as a methodologically best data handling, addressing the limitations outlined above by preserving the uncertainty introduced by missingness and respecting the joint distribution of the covariates (Buuren, 2018). Rather than replacing each missing value with a single summary statistic, MICE generates $m = 5$ completed datasets through iterative conditional imputation, where each variable is imputed sequentially using a regression model conditioned on all other covariates (Buuren, 2018).

QUOTE_VALUE is imputed using predictive mean matching (pmm), which draws imputed values from observed donor values closest in predicted outcome to the missing

unit, thereby preserving the marginal distribution of the variable and avoiding implausible imputed values (Buuren, 2018). RFM_SEGMENT is imputed using polytomous logistic regression (`polyreg`), which is appropriate for unordered categorical variables. All nine covariates are used as predictors in the imputation model to maximise the information available to each conditional model.

In MICE, the imputation model is itself a statistical model that learns the relationships between covariates from the data passed to it. If test-set observations were allowed to contribute to this estimation step, the data leakage would occur. To avoid this, test-set observations are passed to the imputation algorithm with `ignore = TRUE`. This instructs MICE to include those observations in the imputation draws so that they do receive imputed values for their missing covariates but to exclude them from the parameter estimation step. Concretely, the conditional imputation models are fitted on training observations only, and missing values in the test set are subsequently filled in by drawing from those already-fitted models. The test set is thus treated as new, unseen data arriving after training, which correctly mirrors a real deployment scenario in which the imputation pipeline is fixed before any evaluation data are observed (Buuren, 2018).

The $m = 5$ completed datasets produced by MICE are combined into a single dataset. The final imputed value for QUOTE_VALUE is the row-wise mean across the five draws, and the final imputed category for RFM_SEGMENT is the majority vote across the five draws.

CATE Estimators Four causal machine learning estimators are compared. The three meta-learners use random forests as their base learner, implemented via the `ranger` package (Wright and Ziegler, 2017).

Random forests are selected as the base model because they are non-parametric, handle mixed continuous and categorical covariates without requiring distributional assumptions, are robust to irrelevant features, and have been shown to perform competitively in heterogeneous treatment effect estimation benchmarks (Wager and Athey, 2018)(Curth et al., 2024).

The tree parameter is set to 500 as the variance estimation procedures underlying the causal forest require a sufficient number of trees to be reliable (Wager and Athey, 2018).

S-Learner The S-learner trains a single random forest on the full training data, treating the binary treatment indicator W_i as just another covariate alongside $\mathbf{X}_i$ (Künzel et al., 2019). The CATE is then estimated by predicting each test unit’s outcome twice, once under $W_i = 1$ and once under $W_i = 0$ and taking the difference:

$$\hat{\tau}(x)_i^S = \hat{\mu}(1, \mathbf{X}_i) - \hat{\mu}(0, \mathbf{X}_i).$$

T-Learner The T-learner trains separate random forests on the treated ($W_i = 1$) and control ($W_i = 0$) subsamples of the training data (Künzel et al., 2019). The CATE is estimated as:

$$\hat{\tau}(x)_i^T = \hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i),$$

where $\hat{\mu}_1$ and $\hat{\mu}_0$ are the treated and control outcome models. Because each model is fitted on approximately half the training data under balanced randomisation, the T-learner is more susceptible to variance inflation in small samples than the S-learner (Künzel et al., 2019).

X-Learner. The X-learner extends the T-learner through a two-stage cross-fitting procedure (Künzel et al., 2019). In the first stage, outcome models $\hat{\mu}_1$ and $\hat{\mu}_0$ are fitted separately on the treatment and control groups, as in the T-learner. In the second stage, pseudo-residuals are computed for each training unit:

$$\tilde{\tau}(\mathbf{x})_i^{(1)} = Y_i - \hat{\mu}_0(\mathbf{X}_i) \quad \text{for } W_i = 1, \quad \tilde{\tau}(\mathbf{x})_i^{(0)} = \hat{\mu}_1(\mathbf{X}_i) - Y_i \quad \text{for } W_i = 0.$$

Two effect models $\hat{\tau}(\mathbf{x})_1$ and $\hat{\tau}(\mathbf{x})_0$ are then fitted on these pseudo-residuals, and the final CATE estimate is their equally weighted average:

$$\hat{\tau}(\mathbf{x})_i = 0.5 \hat{\tau}(\mathbf{x})_1(\mathbf{X}_i) + 0.5 \hat{\tau}(\mathbf{x})_0(\mathbf{X}_i).$$

Equal weighting is appropriate under balanced randomisation, where the treatment and control groups are of comparable size (Künzel et al., 2019). Treatment assignment follows $W_i \sim \text{Bernoulli}(0.5)$ independently for each unit within each replication, yielding an expected treatment group size of 1000 out of 2000 sampled units.

Causal Forest The causal forest (Wager and Athey, 2018), implemented via the `grf` package, is a purpose-built estimator for heterogeneous treatment effects that differs from the meta-learner approaches in two key respects. First, it uses honest estimation where each tree uses separate subsamples for selecting splits and estimating leaf-level treatment effects, preventing overfitting (Wager and Athey, 2018). Second, it removes the influence of covariates from both the outcome Y_i and the treatment W_i before splitting, so that the tree focuses on treatment effect heterogeneity rather than baseline outcome variation (Wager and Athey, 2018).

3.2 Analysis Approach

To evaluate CATE estimation accuracy and robustness under varying levels of covariate missingness crossed with three missing-data handling strategies, several performance measures are computed across $R = 1000$ simulation replications.

1. CATE Accuracy: RMSE. The primary accuracy metric is the root mean squared error between estimated and true individual treatment effects. Within each replication r , a squared error is first computed per test observation as $(\hat{\tau}(\mathbf{x})_i - \tau(\mathbf{x})_i)^2$, and RMSE is obtained by averaging across all n_{test} test units and taking the square root:

$$\text{RMSE}_r = \sqrt{\frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} (\hat{\tau}(\mathbf{x})_i - \tau(\mathbf{x})_i)^2},$$

where $\hat{\tau}(\mathbf{x})_i$ is the estimated CATE and $\tau(\mathbf{x})_i = p_{1i} - p_{0i}$ is the true individual treatment effect stored from the data-generating process. This yields one RMSE_r per replication per configuration, where a configuration is a unique combination of missingness level, handling method, and CATE estimator. The reported mean RMSE for each configuration is then:

$$\overline{\text{RMSE}} = \frac{1}{R} \sum_{r=1}^R \text{RMSE}_r.$$

Lower values indicate closer agreement between estimated and true individual treatment effects.

2. CATE Stability: SD of RMSE. Stability is measured as the standard deviation of RMSE_r across the $R = 1000$ replications within each configuration:

$$\text{SD}_{\text{RMSE}} = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (\text{RMSE}_r - \overline{\text{RMSE}})^2}.$$

This reflects how consistently an estimator performs across different random samples drawn from the same population. A larger SD indicates that performance varies substantially across realisations of the sampling and missingness process, signalling that the pipeline is fragile and that any single deployment may produce results that differ considerably from the average.

3. Ranking Quality: Top-20% Targeting Overlap. To evaluate whether estimated CATEs produce useful targeting decisions, a top-20% targeting overlap metric is computed within each replication. For each test set, customers are ranked by their estimated $\hat{\tau}(\mathbf{x})_i$ and separately by their true $\tau(\mathbf{x})_i$. The overlap is the fraction of customers in the predicted top 20% who also appear in the true top 20%:

$$\text{Overlap}_{20} = \frac{|\hat{\mathcal{T}}_{20} \cap \mathcal{T}_{20}|}{|\mathcal{T}_{20}|},$$

where $\hat{\mathcal{T}}_{20}$ and $\mathcal{T}_{20}$ are the predicted and true top-20% sets.

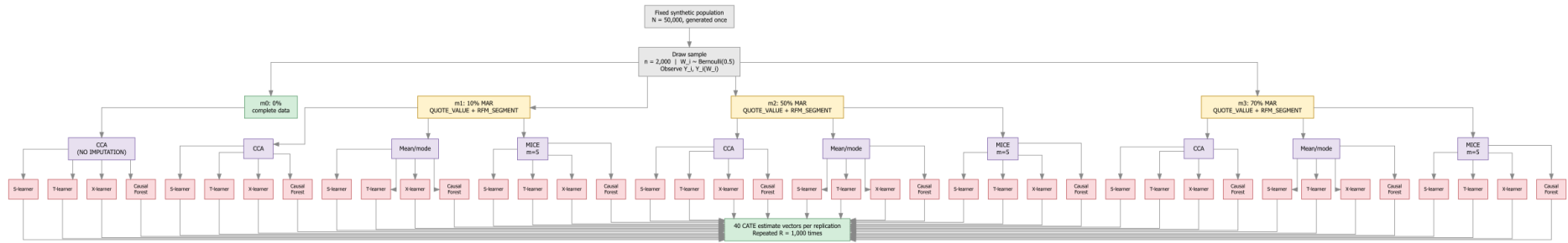


Figure 1: Single replication illustration (all 40 CATE estimation paths)

4 Results

The reference scenario throughout is the complete-data S-Learner (no missingness, no imputation required). A three-way ANOVA was conducted for each diagnosis with missingness level, imputation method, and estimator as factors; all main effects and interactions were statistically significant ($p < .001$) in all cases.

4.1 RQ1: Accuracy of CATE Estimation (RMSE)

Figure 2 presents mean RMSE across all scenarios. The ANOVA shows that the estimator is the dominant source of variance ($SS = 47.94$), far exceeding missingness level ($SS = 1.66$) and imputation method ($SS = 0.89$) (Table 3).

Table 3: Three-Way ANOVA Results for RMSE

term	df	sumsq	meansq	statistic	p.value
missingness_level	3	1.657	0.552	2671.602	0
imputation_method	2	0.890	0.445	2152.983	0
estimator	3	47.935	15.978	77287.053	0
missingness_level:imputation_method	4	0.611	0.153	738.613	0
missingness_level:estimator	9	0.497	0.055	266.956	0
imputation_method:estimator	6	1.145	0.191	923.119	0
missingness_level:imputation_method:estimator	12	0.548	0.046	220.903	0
Residuals	39960	8.261	0.000		

Exact scenario-level RMSE values are obtained from a clustered OLS regression of replication-level RMSE on scenario indicators, with the complete-data S-learner as reference. Full scenario-level SD values are reported in the simulation output files accompanying this thesis.

Figure 2 presents mean RMSE $\pm$ 1 SD across all scenarios. Under complete data, the S-Learner (RMSE = 0.064) and Causal Forest (RMSE = 0.063) achieve the lowest error. The T-Learner (RMSE = 0.138) and X-Learner (RMSE = 0.100) are substantially worse and remain so across all scenarios. Increasing missingness degrades accuracy for all estimators, but the magnitude differs by estimator. At 70% missingness under CCA, the T-Learner reaches RMSE = 0.190 and the X-Learner RMSE = 0.150, while the S-Learner remains at RMSE = 0.059. Among imputation methods, MICE and mean/mode substitution outperform CCA at high missingness for the Causal Forest and S-Learner at 70% missingness, the Causal Forest RMSE under MICE is 0.067 versus 0.089 under CCA.

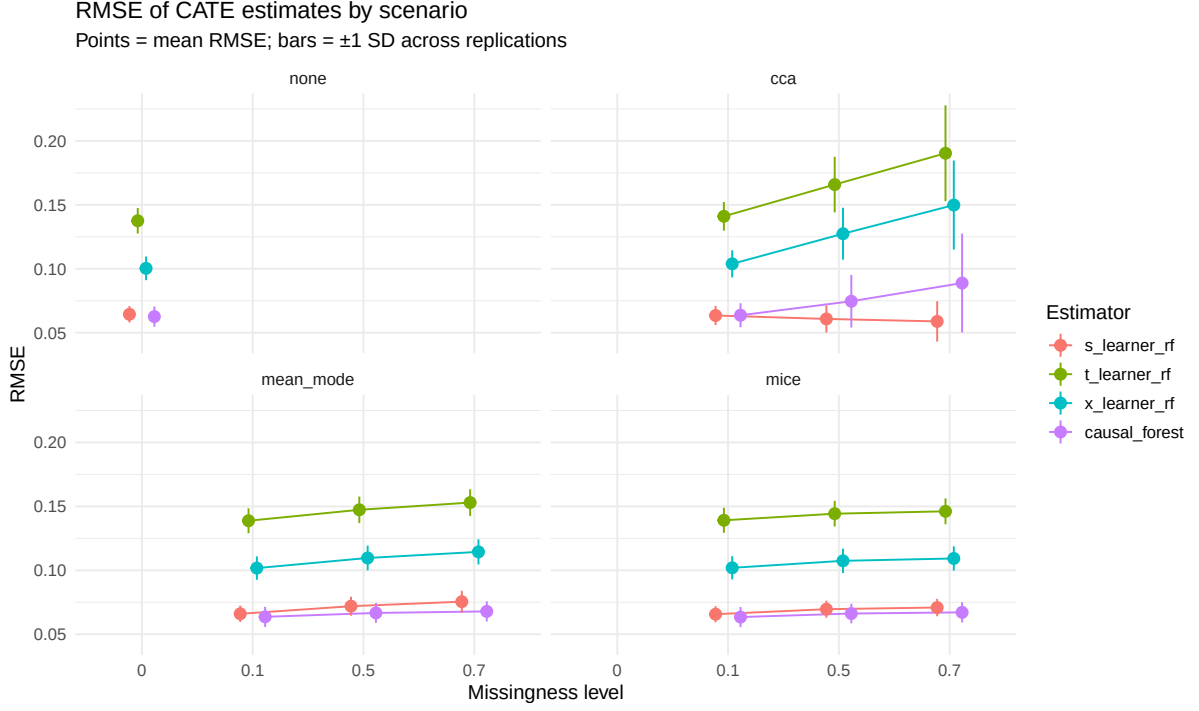


Figure 2: RMSE of CATE estimates by scenario

For the T-Learner and X-Learner, differences across imputation methods are negligible relative to their overall error level.

4.2 RQ2: Stability of CATE Estimation (SD of RMSE)

Stability is measured as the standard deviation of RMSE across the 1000 replications, reflecting how consistently an estimator performs across different random samples from the same population.

At 70% missingness under CCA, the T-Learner SD reaches 0.037, the X-Learner 0.035, and the Causal Forest 0.039. By contrast, under MICE and mean/mode imputation at 70% missingness, the S-Learner (SD = 0.007) and Causal Forest (SD = 0.008) maintain stability comparable to the complete-data baseline. The S-Learner is the most stable estimator overall, with SD < 0.016 across all scenarios including CCA at 70% missingness. Full scenario-level SD values are reported in the simulation output files accompanying this thesis.

4.3 RQ3: Ranking Quality (Top-20% Targeting Overlap)

Figure 3 presents the top-20% targeting overlap — the fraction of customers predicted to be in the top quintile of treatment benefit who are also in the true top quintile. A value of 0.20 corresponds to random selection (dashed line).

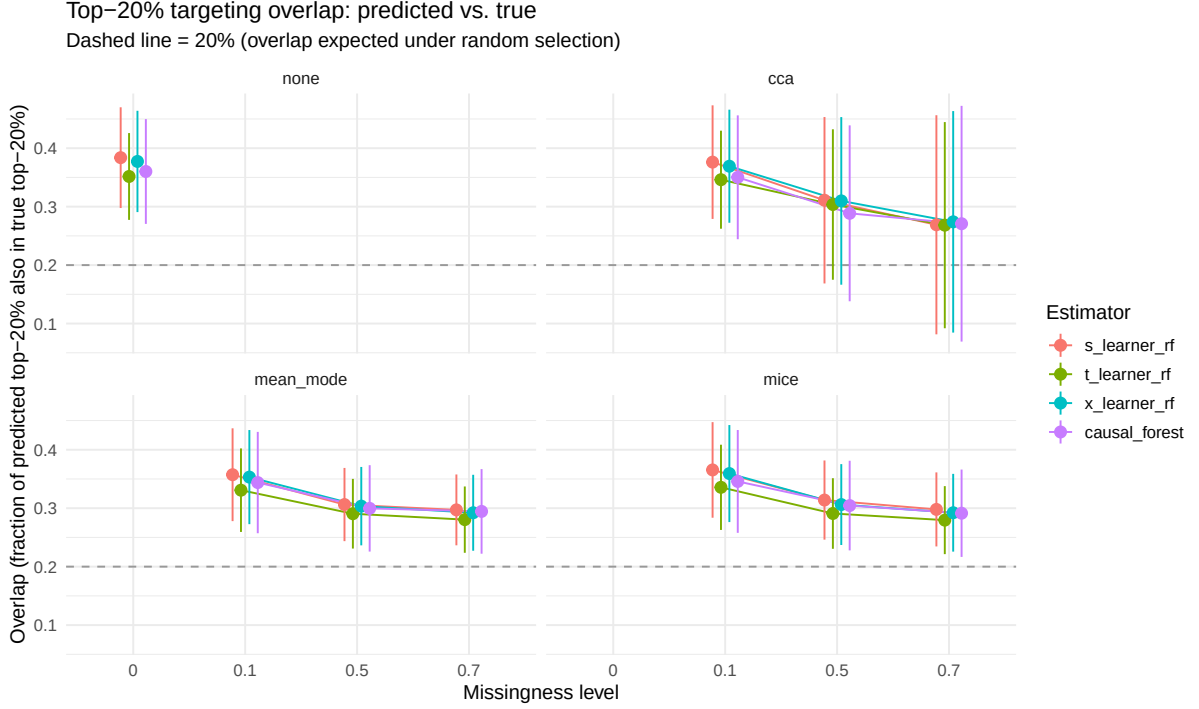


Figure 3: Top-20% targeting overlap by scenario. Points = mean overlap across 1,000 replications; bars = ± 1 SD. Dashed line = 0.20 (random selection baseline).

Under complete data, the Causal Forest and S-Learner achieve the highest overlap (approximately 0.35–0.40), nearly doubling the precision of random selection. The T-Learner performs worst, approaching the random baseline in high-missingness CCA scenarios. The X-Learner is intermediate across all conditions. Under MICE and mean/mode imputation, the Causal Forest and S-Learner remain substantially above the random baseline at all missingness levels. CCA at 70% missingness produces the widest SD bands across replications for all estimators, consistent with the stability findings in RQ2. The ranking quality results closely mirror the RMSE pattern, confirming that accuracy gains translate directly into improved targeting decisions.

5 Discussion

This study examined how covariate missingness, missing-data handling strategy, and CATE estimator choice jointly affect the accuracy, stability, and ranking quality of heterogeneous treatment effect estimation in a synthetic CRM setting. The discussion situates findings within two primary theoretical backbones: the benchmarking literature on ITE methods (Röbber and Schoder, 2022) and the causal treatment of missing data in treatment effect estimation (Berrevoets et al., 2023), supplemented by the missing data imputation comparison literature (Camille et al., 2002).

5.1 RQ1: Accuracy of CATE Estimation

Interpretation of Findings The dominant finding is that estimator choice accounts for the overwhelming majority of variance in RMSE ($SS = 47.94$), far exceeding the contributions of missingness level ($SS = 1.66$) and imputation method ($SS = 0.89$), meaning that it’s the choice of estimator model that determines the final ”best” estimate. Another highlight is the slight RMSE decrease of the S-Learner under CCA at 70% missingness, reflecting a selection effect. More specifically, listwise deletion at severe missingness does not remove records at random. Because missingness in the simulation depends on CHANNEL_TYPE and DEAL_OWNER, the records that survive deletion are a specific, non-random subset of customers who happen to be more similar to each other than the full population. A single flexible model such as the S-learner finds it easier to fit a homogeneous group, so its RMSE decreases, not because estimation quality improved, but because the problem itself became artificially easier.

Comparison to Literature The dominance of estimator choice over missing data conditions is consistent with Rößler and Schoder (2022), who demonstrated across 27 synthetic and 6 real-world datasets that performance differences between methods are substantially larger than differences attributable to data characteristics. The near-equivalence of the S-Learner and Causal Forest aligns with Rößler and Schoder (2022)’s finding that the S-Learner exhibits good predictive power, while the poor performance of the T-Learner is consistent with Künzel et al. (2019)’s theoretical observation that two-model approaches perform poorly when treatment effects are simple relative to outcome variance, precisely the case in the present heterogeneity data-generating process.

Theoretical Implications The finding that estimator architecture dominates missing data conditions has an important theoretical implication. The quality of the estimation procedure itself is a more consequential design choice than whether MICE or mean substitution is used upstream. Furthermore, Berrevoets et al. (2023) show that in settings where missingness and treatment are causally entangled, the question of *what* to impute matters as much as *how* to impute. The present study complements this by showing that even in the simpler MAR setting, *which estimator* processes the imputed data is more consequential than the imputation method itself. Together, these findings suggest that future theoretical work should examine the joint properties of imputation and estimation procedures rather than treating them as separable design choices.

Managerial Implications For CRM practitioners, the practical message is clear that analysts should invest time in exploring and learning about estimation models. An organisation that switches from a T-Learner to an S-Learner or Causal Forest will achieve larger accuracy gains than one that upgrades from mean/mode imputation to MICE

while retaining a T-Learner. At low missingness rates (10%), imputation method is essentially irrelevant - a finding that provides reassurance to organisations with moderate data quality limitations. Only when missingness exceeds 50% does imputation strategy meaningfully affect outcomes, and even then, the gains from MICE over mean/mode imputation are modest for the best-performing estimators.

5.2 RQ2: Stability of CATE Estimation

Interpretation of Findings Stability, measured as the SD of RMSE, was low and comparable across all estimators under complete data and low missingness ($SD \leq 0.010$). CCA at high missingness dramatically increased replication-to-replication variability: the T-Learner reached $SD = 0.037$, the X-Learner $SD = 0.035$, and the Causal Forest $SD = 0.039$ at 70% missingness. MICE and mean/mode imputation eliminated this instability by preserving the full sample, with the S-Learner ($SD = 0.007$) and Causal Forest ($SD = 0.008$) maintaining near-baseline stability even at 70% missingness. The S-Learner was the most stable estimator overall across all conditions.

Comparison to Literature The instability observed under CCA at high missingness directly reflects the mechanism described by Camille et al. (2002) where listwise deletion reduces the analytic sample, resulting in fewer degrees of freedom and larger standard errors. In the present repeated-sampling framework, this translates into substantially higher variance in RMSE across replications when the effective sample fluctuates around 600 observations following 70% listwise deletion from 2000 observations. The high volatility of multi-stage estimators under CCA also connects with Rößler and Schoder (2022), who identified volatility as a pervasive problem across ITE methods and noted it was particularly pronounced for the two-model approach, in this study the T-Learner.

The instability finding also connects to the broader argument of Berrevoets et al. (2023) that missing data in treatment effect settings “should be treated differently” from standard supervised learning. Their sensitivity analysis shows that as missingness increases, the trade-off between prediction gain from imputation and bias reduction shifts at very high missingness, the patterns causing bias become less apparent, producing counterintuitive stability effects. The present study observes a related phenomenon. The S-Learner becomes more stable at high missingness under CCA due to the homogeneity of the retained sample, suggesting that missingness itself can change the effective difficulty of the estimation problem.

Theoretical Implications The stability results reveal a mechanism where the combination of a multi-stage estimator and listwise deletion at high missingness creates compounding instability, where both effective sample size and training signal quality vary substantially across replications of the missingness process.

Managerial Implications An organisation deploying a T-Learner with CCA at 70% missingness cannot reliably interpret any single set of CATE estimates. This makes prioritisation decisions potentially counterproductive. The practical recommendation is unambiguous. If CRM data is severely incomplete, listwise deletion should be avoided in favour of any imputation strategy that preserves the full sample. Even mean/mode substitution reduces instability to levels comparable to the complete-data baseline for the best-performing estimators.

5.3 RQ3: Ranking Quality

Interpretation of Findings The top-20% targeting overlap is closely similar to the RMSE pattern across all conditions. The S-Learner and Causal Forest achieved overlaps of approximately 0.35-0.40 under complete data and low missingness, nearly doubling the precision of random selection.

Comparison to Literature The finding that RMSE performance translates directly into targeting quality is consistent with Rößler and Schoder (2022), who demonstrated that the choice of ITE method substantially affects targeting policy outcomes and that methods with similar average performance can differ substantially in targeting utility. Their Qini curve analyses showed that in datasets with small ATEs similar to the modest individual treatment effects in the present study, even well-performing methods struggled to substantially outperform random targeting, mirroring the ceiling effect observed here.

The difficulty of improving targeting quality under missing data also connects to Berrevoets et al. (2023), who show empirically that imputing the wrong variables or imputing all variables when some missingness is informative of treatment can yield CATE estimates worse than no imputation at all. While the present MAR setting does not exhibit MCM, the underlying insight holds that the interaction between missing data handling and estimator choice determines whether targeting decisions are based on genuine treatment effect signal. The T-Learner’s near-random targeting overlap under high-missingness CCA demonstrates this.

Managerial Implications A near-doubling of targeting precision at scale, translates into substantially more efficient allocation of follow-up sales effort. However, organisations should note that this advantage deteriorates sharply under CCA at high missingness. The T-Learner’s targeting overlap approaches the random baseline at 70% missingness with listwise deletion, meaning its predictions carry no useful information for prioritisation. Organisations relying on such analyses may be allocating sales resources at random while believing they are acting on analytical insight.

5.4 Additional Analysis: Why Is Overall Performance Weak?

A notable feature of the results is that targeting overlap remained modest in absolute terms (approximately 0.35-0.40 versus 1 maximum) even for the best-performing estimators under complete data. To understand this, the total MSE for each estimator can be decomposed into two additive components:

$$\text{MSE} = \underbrace{(\bar{\hat{\tau}} - \bar{\tau})^2}_{\text{ATE bias}} + \underbrace{\mathbb{E}\left[\left((\hat{\tau}_i - \bar{\hat{\tau}}) - (\tau_i - \bar{\tau})\right)^2\right]}_{\text{heterogeneity error}},$$

where $\bar{\hat{\tau}} = \frac{1}{n} \sum_i \hat{\tau}_i$ is the mean estimated treatment effect and $\bar{\tau} = \frac{1}{n} \sum_i \tau_i$ is the mean true treatment effect. The first term captures whether the estimator correctly recovers the average lift across all customers. The second term captures whether the estimator correctly ranks customers by their individual deviation from that average - the main task in CATE estimation. A model can have negligible ATE bias while still performing poorly on the second term if it assigns similar $\hat{\tau}_i$ to customers whose true τ_i varies substantially.

This decomposition follows from the standard variance decomposition of squared error (Curth et al., 2024). Letting $a = \hat{\tau}_i$ and $b = \tau_i$, the identity $\mathbb{E}[(a - b)^2] = (\mathbb{E}[a] - \mathbb{E}[b])^2 + \mathbb{E}[((a - \mathbb{E}[a]) - (b - \mathbb{E}[b]))^2]$.

From the Table 4), between 84% and 97% of total MSE across all estimators originates the failure to accurately recover individual-level variation in treatment effects rather than from ATE bias. The ATE bias component is negligible and near-identical across all four estimators.

Table 4: MSE decomposition by estimator, averaged across all scenarios.

Estimator	ATE Bias (MSE)	Het. Error (MSE)	Total MSE	% Het. Error
S-Learner	0.000663	0.00353	0.00419	84.4%
T-Learner	0.000576	0.01840	0.01900	97.1%
X-Learner	0.000549	0.00961	0.01020	94.9%
Causal Forest	0.000586	0.00340	0.00398	87.6%

This pattern is consistent with Rößler and Schoder (2022)’s observation that methods with good average performance do not necessarily produce useful targeting curves in datasets with small (C)ATEs, and with the theoretical limits of heterogeneous treatment effect estimation discussed in the causal machine learning literature (Berrevoets et al., 2023) that when the true treatment effect varies within a narrow, bounded range as in the present data-generating process where $\tau(x)_i \in [0, 0.10]$, even correctly specified estimators in the perfect world face limit in achievable heterogeneity recovery set by the signal-to-noise ratio of the treatment effect variation itself rather than by the estimation procedure.

A further implication concerns the practical value of CATE estimation in CRM environments with genuinely small treatment effects. Berrevoets et al. (2023) argue that creating standardised guidelines for handling missing data in treatment effect settings is difficult without deeper theoretical understanding of the causal structure of the missingness. The present study reinforces this from an applied angle. Even under the relatively clean MAR setting with well-understood missingness mechanisms, estimation quality is ultimately bounded by the magnitude of the treatment effect heterogeneity in the data-generating process. Practitioners should therefore invest both in data quality improvement by reducing missingness in key covariates and in realistic assessment of the degree of treatment effect heterogeneity in their context before committing to complex CATE estimation pipelines.

6 Conclusion

This thesis set out to investigate a problem of when CRM data is incomplete, how much does it matter, and for what? From the Introduction and Discussion it has been made clear that analytical recommendations are only as reliable as the data and pipelines that produce them. This study examined that reliability systematically, in the context of Conditional Average Treatment Effect estimation for customer targeting in a B2B CRM environment.

Summary of Main Findings. Across 1000 replications of a factorial simulation combining four missingness levels, three imputation methods, and four CATE estimators, three main findings emerge. First, estimator choice is the dominant driver of CATE estimation quality. The S-Learner and Causal Forest consistently outperformed the T-Learner and X-Learner across all conditions. Second, missing-data handling strategy matters, but primarily at high missingness levels and only for the best-performing estimators. Complete-case analysis at 70% missingness dramatically degraded both accuracy and stability for all estimators, while MICE and mean/mode imputation maintained near-baseline performance by preserving the full sample. Third, all estimators recovered the average treatment effect with comparable accuracy. The critical differentiator was the ability to rank individuals by treatment benefit. The modest absolute targeting overlap achieved even by the best estimators reflects a limitation imposed by limited treatment effect heterogeneity in individual treatment effect.

Limitations. Several limitations should be acknowledged. The simulation is calibrated to a single B2B CRM context in manufacturing sector. Furthermore, the study considers only MAR missingness. MNAR mechanisms are plausible in CRM environments where low-engagement customers are systematically less likely to have complete records, and

would be expected to more severely compromise causal identification. The study does not consider the more complex missingness structures described by Berrevoets et al. (2023), where missingness and treatment assignment are causally entangled. This is scenario likely to arise in observational CRM data where sales associates may record information selectively based on their treatment (follow-up) decisions. Missingness was imposed on only two covariates, whereas real CRM datasets may exhibit broader multivariate missingness patterns. Finally, the imputation methods evaluated represent a subset of available strategies. Finally, the data-generating process is relatively simple.

Recommendations for Future Research. Future research should examine CATE estimation under MNAR missingness and, critically, under the MCM framework of Berrevoets et al. (2023), where missingness is causally entangled with treatment assignment. This represents the most important theoretical gap between the present study and real-world CRM observational data, which would also mean that data-generation-process becomes far more complex.

Appendices

Appendix A Covariate Distributional Parameters

Table 5 reports the distributional parameters used to generate each synthetic CRM covariate. These parameters were chosen to approximate the empirical distributions observed in the reference CRM project while remaining fully synthetic and containing no company-specific confidential values.

Table 5: Distributional parameters for synthetic CRM covariates

Variable	Distribution	Parameters
DEAL_OWNER	Uniform categorical	8 levels, equal probability (1/8 each)
CHANNEL_TYPE	Categorical	10 levels; weights: REF 0.10, A 0.01, B 0.04, C 0.22, D 0.05, E 0.18, F 0.03, G 0.03, H 0.12, I 0.22
QUOTE_REQUEST_MONTH	Categorical	Months 1–12; spring peak: Jan 0.06, Feb 0.07, Mar–Apr 0.11, May–Jun 0.09, Jul 0.08, Aug–Sep 0.09–0.10, Oct 0.09, Nov 0.06, Dec 0.05
NUM_PREVIOUS_ORDERS	NegBin, clipped	$\mu = 7.7$, size = 0.15; clipped to [0, 150]
CUSTOMER_LIFETIME_VALUE_LOG	Normal	$\mu = 7.5 + 0.35 \log(1 + \text{NUM_PREV_ORDERS})$, $\sigma = 0.55$
RFM_SEGMENT	Derived categorical	Score = $0.60 \log(1 + \text{PREV_ORDERS}) + 0.50 z(\text{CLV_LOG}) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.36)$; cut into 11 equal-frequency segments
NUM_PRODUCT	NegBin, clipped	$\mu = 4$, size = 0.5; minimum 1
QUOTED_VALUE	Log-Normal	$\mu_{\log} = 7.6 + 0.08 \text{NUM_PRODUCT}$, $\sigma_{\log} = 0.45$
DAYS_TO_SEND_QUOTE	NegBin, clipped	$\mu = 10$, size = 0.5; clipped to [1, 300]
DAYS_BETWEEN_FIRST_AND_LAST_QUOTE_LOG	Mixed	60% zero-inflated; non-zero: NegBin($\mu = 150$, size = 0.3), clipped at 0; log1p-transformed

Appendix B Treatment Effect Weight Parameters

Table 6: RFM segment weights (γ^{RFM}) and deal owner weights ($\gamma^{\text{DEAL-OWNER}}$) used in the treatment effect generating process

RFM Segment	γ^{RFM}	Deal Owner	γ^{owner}
REFERENCE	0.00	REFERENCE	0.00
A	0.15	A	0.10
B	0.10	B	0.08
C	0.05	C	0.05
D	0.00	D	0.03
E	0.00	E	-0.03
F	-0.05	F	-0.05
G	-0.08	G	-0.08
H	-0.10		
I	-0.15		
J	-0.20		

Appendix C Git

<https://github.com/martab-coder/msc-thesis.git>

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